St Mary's DSG, Kloof

Grade 10	Physical Science Cycle Test Electricity	24 August 2018
TIME: 55 min		MARKS: 60

ASSESSOR: G. Main MODERATORS: K. Bolton and T. Kirsten

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 6 pages.
   A data sheet is provided.
   Please check that your question paper is complete.

2. Answer on your own foolscap paper and write your Physical Science teacher's name on your paper.

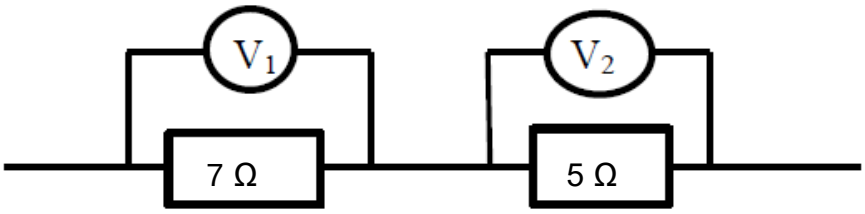
3. Staple your question paper to the back of your answer paper.

Question 1 Multiple Choice

1.1 A voltmeter is connected in parallel with a resistor in an electric circuit as it has

A a low resistance and measures potential difference
B a high resistance and measures work done per unit charge
C a low resistance and measures the rate of flow of charge
D a high resistance and measures the number of charges flowing through it each second

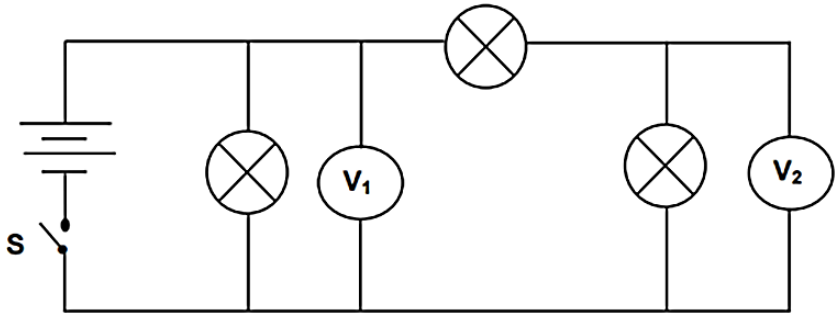
1.2



Consider the above diagram; if a current is flowing through both resistors;

A V_1 will have a higher reading than V_2
B V_2 will have a higher reading than V_1
C the reading on V_1 and V_2 will be the same
D potential difference is inversely proportional to resistance.

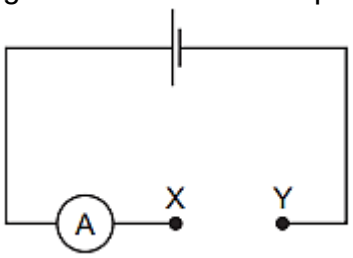
1.3 Three **identical** light bulbs are connected in a circuit as shown below.



The resistances of the battery and connecting wires can be ignored.
Which ONE of the following statements is CORRECT when switch **S** is closed? The reading on V_1 is …

A half that on V_2.
B equal to that on V_2.
C twice that on V_2.
D three times that on V_2.

1.4 The diagram shows an incomplete circuit.



Four wires of different length and thickness are connected in turn between point X and point Y. All four wires are made of the same metal. Which wire will cause the **greatest** reading on the ammeter?

A long and thick
B long and thin
C short and thick
D short and thin

[4 x 2 = **8**]

Question 2

The potential difference across a glowing bulb is 220 V.

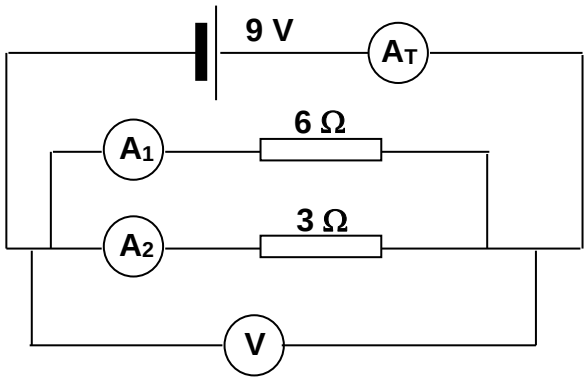
2.1 Define *current*. (2)

2.2 A charge of 3,5 C flows through the filament of the bulb in 8 min.

2.2.1 Calculate the electric current passing through the filament. (3)

2.2.2 Calculate the amount of energy converted to heat and light. (3)
[8]

Question 3



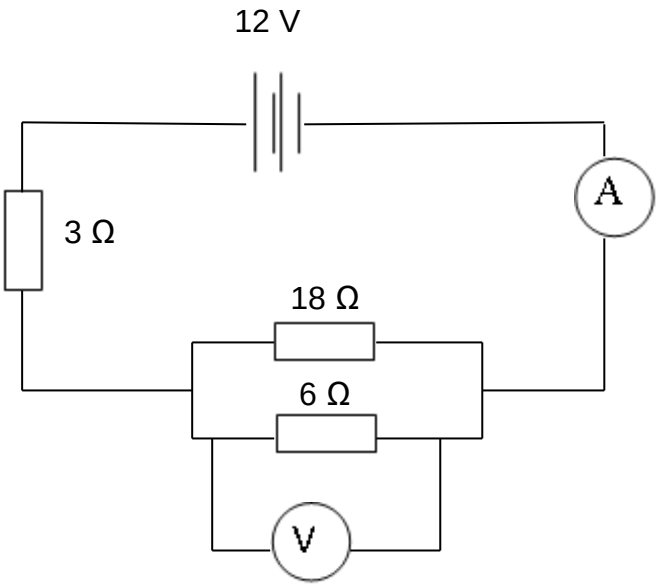
For the above circuit, given $\mathbf{A_T}$ = 4,5 A, calculate the following:

3.1 Total resistance. (3)
3.2 The reading on voltmeter V. (2)
3.3 The readings on A_1 and A_2. (4)
3.4 The work done in moving 3 Coulombs of charge through the 3 Ω resistor. (3)

[12]

Question 4

A 12 V battery is connected in a circuit as shown in the diagram below.

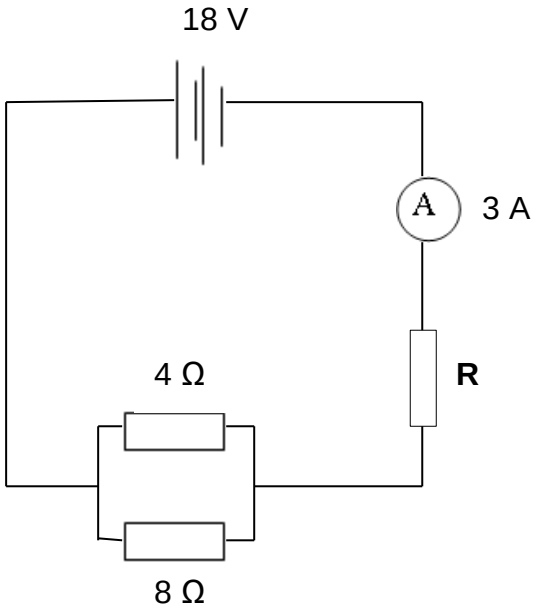


Calculate:

4.1 the resistance of the parallel combination (3)
4.2 the total resistance of the given circuit (1)
4.3 the reading on the ammeter (3)
4.4 the reading on the voltmeter (3)

[10]

Question 5



5.1 What is the potential difference across the parallel portion of the circuit? (5)

5.2 What is the potential difference across the resistor R? (2)

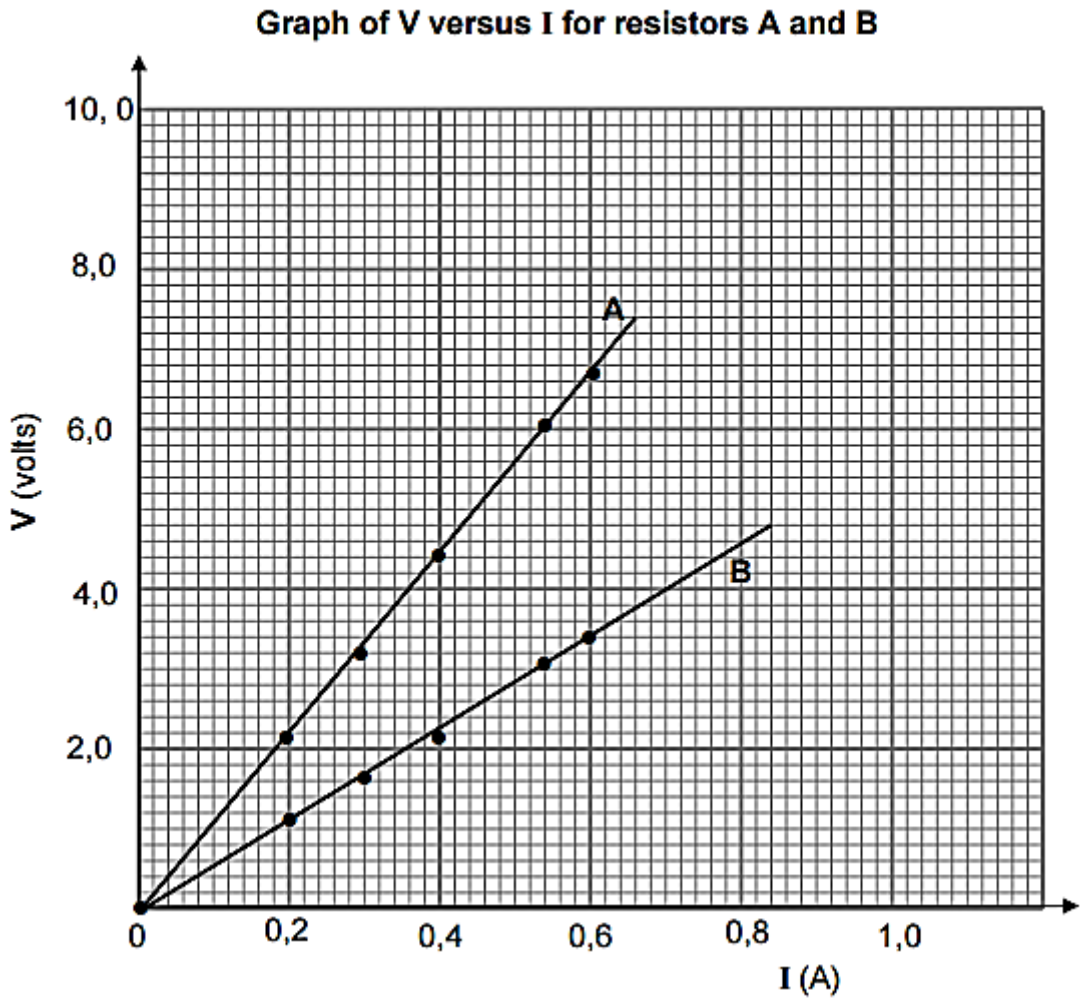
5.3 What is the resistance of R? (2)

5.4 How much energy is used up when the current flows through resistor R for 35 s? (3)

[12]

Question 6

Learners want to construct an electric heater using one of two wires of the same material, A and B, of different resistances. They conduct experiments and draw the graphs as shown below.



6.1 Apart from temperature, write down TWO other factors that the learners should consider to ensure a fair test when choosing which wire to use. (2)

6.2 Calculate the resistance of Wire A and Wire B. (5)

6.3 Assuming all other factors are kept constant, state which ONE of the two wires will be the most suitable to use in the heater. Use suitable formulae to show clearly how you arrive at the answer.

(3)
[10]